WHEN DOES EXPERIMENTAL KNOWLEDGE IMPROVE SCIENTIFIC DECISIONS?

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ABSTRACT

Autonomous scientific agents may improve predictions without producing knowledge that reliably guides new decisions. We study this distinction in ChemWorld through controlled initial descriptions, executable knowledge artifacts and complete action plans. Two model configurations each entered 135 scheduled campaigns across 45 matched task–world clusters. Prediction errors decreased on average, but the registered selective-correction criteria were not met, and executable summaries retained different amounts of predictive information. In a separate DeepSeek cohort, submitted laws selected the optimum in 0/45 scheduled unseen-plan cases, compared with 11/45 participant choices. We then held public evidence fixed in a factorial intervention across ten new worlds, crossing model-generated or numerically fitted quadratic laws with fresh-agent or deterministic selection. All 160 condition slots completed. Under the same maximizer, fitting changed mean regret by -0.00538 (95% interval [-0.01630, 0.00061]), failing the prespecified material-benefit criterion. Fresh agents and the maximizer made identical choices in all 40 fitted-law pairs, while a nearest-evidence baseline had similarly low regret. These results limit a general inability-to-use-laws account: historical law/action disagreement coexists with high agreement on a simpler fixed-evidence surface. They establish bounded diagnostic findings, without isolating the cause of differences between protocols or demonstrating a general repair method or transfer to new physical conditions.

1 INTRODUCTION

Autonomous scientific agents choose experiments, interpret observations and recommend what to do next. The value of experimental knowledge therefore depends on the decisions it supports. A useful outcome, an accurate prediction and an executable scientific summary answer different questions: an agent may find a productive recipe, predict a local response, or express a relation without being able to select a good plan under new conditions.

This problem matters as language-model agents operate chemistry tools and self-driving laboratories (Boiko et al., 2023; Bran et al., 2024; Szymanski et al., 2023; Darvish et al., 2025; Song et al., 2025; Vriza et al., 2026). Interactive discovery environments make repeated experimentation accessible (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Zheng et al., 2026; Yang et al., 2026; Batzoglou, 2026). Predict-then-optimize and decision-focused learning already establish that predictive error and downstream decision loss can differ (Elmachtoub & Grigas, 2022; Wilder et al., 2019). The additional question for a scientific agent is how autonomously acquired evidence, a supplied prior and a submitted knowledge artifact relate to the actual decision reached by the complete system.

We use ChemWorld to make these objects separately observable. Within a matched cluster, the external world, public operations and resources remain fixed while the supplied initial description is opaque, aligned or misspecified at a declared entity, parametric or structural locus (Qiu et al., 2026). A persistent session performs experiments and submits predictions and typed laws. Independent evaluators score those artifacts and execute complete action plans. The assignment changes participant-facing information; it does not directly manipulate an unobservable internal belief. The foundation paper establishes the bounded environment and replay semantics, while this study evaluates complete agent–tool configurations.

Figure 1: **Endpoint success does not reveal what the agent learned.** ChemWorld varies the supplied initial description while fixing the executable world, public rules, and budget. Held-out predictions, executable knowledge, and unseen-plan decisions are separate readouts. The diagram describes the design; its arrows do not identify internal belief or causal mediation through submitted knowledge.

The primary study used a fixed DeepSeek-v4-flash experimental-agent configuration, with 135 scheduled cells nested within 45 task–world clusters. An independent GPT-5.6-sol successor used the same public scientific surface. Matched-evidence controls, unseen-plan selection and failure-aware information strategies distinguish prediction, artifact fidelity and actual choice. We then test a proposed intervention directly: hold public evidence fixed and cross the source of a quadratic law with the rule selecting a plan. This ten-world factorial block finds no supported material benefit of fitted-law replacement and high agent/maximizer agreement. Together, the studies locate observed gaps and provide a boundary to a universal law-use failure account. Different protocols are not randomized against each other; neither their contrast nor the factorial intervention identifies internal psychological mediation.

2 Environment and Evaluation

2.1 ChemWorld as a controlled causal probe

ChemWorld separates three objects. The executable world contains hidden state, transition and measurement laws, resources, and terminal assays. The public task exposes typed operations, instruments, observations, and an initial model. The evaluator owns held-out truth, replay, and scoring. For world w and initial-model arm a, the hidden transition process remains fixed while the supplied model $M_{0,a}$ changes. The aligned model is correct at one declared locus, the misspecified model is explicit but wrong at that locus, and the opaque arm withholds the corresponding structure. The design changes participant-facing information while holding world physics and the available operation and measurement semantics fixed.

Actions are transactions, not free-form prose. The host validates a typed request, checks resources and preconditions, commits the state transition, and returns a public observation or a structured rejection. Failed actions and discards remain in the trajectory. Every submitted action, commit, rollback, measurement, assay, and resource delta is recorded for exact replay. These semantics matter: an action recommendation is scientific evidence only when the evaluator can execute the same complete plan without filling in hidden workflow choices.

2.2 Observable outcomes and measurement limits

We distinguish five downstream outcomes. **Evidence acquisition** asks which informative experiments are selected and observed. **Numerical revision** asks whether held-out counterfactual prediction error falls. **Structural identification** asks whether the agent rejects a false form and recovers the registered relation. **Executable compression** asks whether a typed law preserves the information in conditional predictions. **Decision transfer** asks whether the final state selects a good previously unseen plan. A sixth requirement, **evaluator validity**, asks whether the control and metric actually measure the decision estimand. Success at one stage is neither defined as nor assumed to imply success at the next.

At five checkpoints per session, the participant reports initial-model reliability, predictions, uncertainty, evidence references, an executable law, and the next experimental intent. The evaluator executes prespecified counterfactual query sets independently of the participant. The DeepSeek-v4-flash surface scores 675/675 checkpoints from 420/420 truth executions; the GPT-5.6-sol surface scores 669/675 checkpoints from its own 420/420 truth executions. The evaluator later runs each available final typed law on the same coordinates. Paired blind replay evaluates the final recommendation against the observed incumbent, while a separate longitudinal assay reveals eight outcome-hidden complete ActionPlans only after autonomous exploration has ended.

3 Experimental programme

The primary DeepSeek cohort comprises 135 separate sessions nested within 45 independent task–world clusters. The registered improvement contrast depends on initial error headroom; a failed gate does not establish an absence of correction ability. Initial predictions and contradicted relations provide its manipulation-check context.

The prospective programme is layer-stratified. Entity interventions cover five task families and five independently selected public worlds per task. Parametric and structural blocks each cover two validated task families and five worlds. Every task–world cluster contains opaque, aligned, and misspecified arms, yielding 135 separate sessions nested within 45 independent task–world clusters. Campaign length is locus-specific: eight, ten, or twelve complete experiments, with five checkpoints in every session. Exploratory, validation, prospective, matched-evidence, and open-action worlds remain separated. We abbreviate the prospective cohort as C2, the matched partition-packet diagnostic as B2, its typed-law/action control as B3, and the entity/parametric/structural loci as A-E/A-P/A-S.

The primary contrast tests selective evidence-driven correction. If $E^{(\ell)}_{a,k}$ is held-out error for arm a, checkpoint k, and locus ℓ, then

$$C_\ell = \left(E^{(\ell)}_{\mathrm{mis},0} - E^{(\ell)}_{\mathrm{mis},K}\right) - \left(E^{(\ell)}_{\mathrm{aligned},0} - E^{(\ell)}_{\mathrm{aligned},K}\right).$$

Success requires greater correction in the misspecified arm, improvement of that arm, and no material deterioration of the aligned arm. Loci are decided separately; unlike intervention semantics are not pooled. Failed cells stay in the scheduled denominator; confirmatory correction gates use adverse bounds, while last-observation and zero-improvement imputations are sensitivities. Only infrastructure failures without a persisted trajectory can resume under a fixed attempt cap.

Matched-evidence sessions use cloned worlds and give every arm the same counterevidence after a pre-response. They reveal conditional post-packet updating but, without a turn-matched no-packet control, do not identify a pure evidence-packet effect. The longitudinal action matrix separately contains three tasks, five worlds, and three arms (45 scheduled cells). After 12 autonomous experiments, each agent ranks eight new plans; regret and Top-1 are primary action readouts, while complete-rank correlation and law adequacy are diagnostics. Table~1 keeps these layers and their claim boundaries explicit.

Provenance is block-specific: C2 and B3 have matched scheduled DeepSeek-v4-flash-high and GPT-5.6-sol-medium surfaces; matched evidence adds complete denominators and a DeepSeek-low B2 reasoning-budget ablation; the four-condition successor retains model-specific donor eligibility. Historical controls and model configurations retain separate denominators. The new factorial block instead prespecifies an equal-weight mean over models and repeats within world.

4 Results: from priors to laws

4.1 Prior-conditioned search and unmet correction criteria

All 135 scheduled sessions produced final records. Participants completed 1,243/1,260 planned experiments; 121 sessions met operational eligibility. The denominator retains 26 discarded lifecycles, 13 resource-ledger rejections, and all right-censored cells. Every session submitted five checkpoints, providing 6,300 counterfactual predictions and 24,300 query–metric values.

Table 1: **Executed evidence layers and claim boundaries.** A completed work package may contain a retained scientific rejection; unstarted units are not silently removed.

Layer	Units	Execution	Supported role
Prospective C2	270 scheduled	DeepSeek 121 complete; GPT 126 complete	Search, prediction, law, incumbent replay
Matched evidence	75 sessions	DeepSeek high + GPT medium (60); DeepSeek low (15)	Conditional numerical–exact-law-expression dissociation
Identifiable-law B3	60 scheduled	DeepSeek 17+13 failures; GPT 30	Structural recovery and action bridge
Action assays	45 cells + 360 slots	Open action + four conditions/model	Descriptive and failure-aware action transfer
Factorial intervention	160 slots; 10 worlds	120/120 sessions; 200/200 physics/replays	Fixed-evidence law and decision-rule replacement
Evaluator controls	16 unit versions	Provider-free; original stops retained	Rank validity versus action validity

Arm assignment was reflected in behavior. The first complete recipe differed between aligned and misspecified cells in 45/45 matched clusters, between opaque and aligned in 45/45, and between opaque and misspecified in 44/45. This is a manipulation check rather than a causal effect estimate because there are no repeated same-arm sessions. Search continued after the first proposal: 91.2% of completed experiments used a unique recipe, 84.4% of session optima appeared after the midpoint, and 32.6% appeared in the last completed experiment.

Correct-prior utility was task dependent. In entity-level partition, the aligned arm showed a +0.200 best-endpoint advantage over the misspecified arm in 5/5 worlds. In structural crystallization, a +0.141 first-experiment head start narrowed to +0.055 as the disadvantaged arm explored. In structural partition, aligned and misspecified descriptions both helped relative to opaque identifiers while differing little from one another. The supplied-model arms therefore occupied different search landscapes, without identifying a stochastic participant effect or imposing one endpoint ordering.

Prediction error nevertheless fell on average in every arm at every locus. Reductions for opaque, aligned, and misspecified cells were 0.111/0.097/0.097 at the entity locus, 0.090/0.033/0.065 at the parametric locus, and 0.219/0.228/0.221 at the structural locus. The stricter selective-correction contrasts were -0.214 for entity ($p = 0.990$), +0.033 for parametric ($p = 0.079$), and -0.224 for structural ($p = 1.000$); none passed. General predictive learning did not become preferential correction of a wrong starting model (Fig.~2).

4.2 MATCHED PACKETS EXPOSE NUMERICAL–EXPRESSION DISSOCIATION; B3 TESTS STRUCTURE

Matched packets provide conditional evidence response, with the packet and extra response turn bundled. In the parametric block, all five DeepSeek misspecified summaries rejected the supplied high-potential direction. B2 produced low post-packet error but no exact wrong-arm law expression (0/5 in each model and DeepSeek-low); its one-pair linear/power alias makes it an underidentifying surface. These observations cannot establish internal structural-identification failure.

B3 instead uses reference-fitter-qualified multi-pair evidence. It retained 30 GPT completions and 17 DeepSeek completions plus 13 schema failures. Joint recovery was 5/30 versus 0/30, and scheduled useful gain was 0/18 for both. The contrast is bounded by differential availability and the specified function family. Full packet, reasoning-budget and B3 results, including every failure denominator, appear in the appendix (Fig.~6).

4.3 EXECUTABLE LAWS LOSE INFORMATION FROM PREDICTIONS

All 135 DeepSeek laws executed, but 84 lost information relative to final explicit predictions; mean law MAE was 0.237 and compression loss 0.069. Legal full-basis controls reproduced 135/135 prediction states at mean MAE 4.25×10^{-13}, localizing the gap to participant distillation rather than typed-interface capacity. Blind incumbent replay completed 726 executions for 121 cells, with recommendations better/equivalent/worse in 1/119/1.

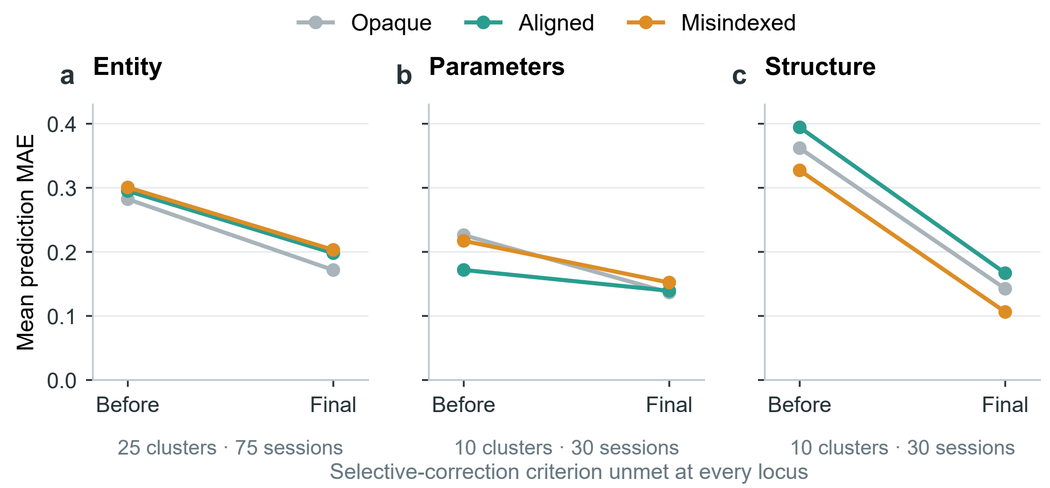

Figure 2: **Prediction improves, but the evidence does not establish selective repair of the wrong model. a–c,** Mean pre-evidence and final errors by arm in entity, parametric, and structural blocks; lines connect aggregate means, not individual trajectories or confidence bounds. The registered selective-correction criteria are unmet (one-sided $p = 0.990, 0.079, 1.000$), and initial error limits improvement headroom. First recipes differ in 45/45 aligned–misindexed, 45/45 opaque–aligned, and 44/45 opaque–misindexed clusters; this retrospective check has no repeated same-arm baseline.

The matched GPT-5.6-sol surface retained 126 completed, 3 failed, and 6 right-censored cells. All locus gates again failed. GPT-5.6-sol versus DeepSeek-v4-flash law MAE was 0.1753 (129 laws) versus 0.2371 (135 laws), and compression loss was 0.0142 versus 0.0686; blind gain was -0.0001 (126 cells) versus -0.0010 (121 cells). Lower observed compression error therefore coexisted with near-zero blind gain (Fig.~3); this matched cross-configuration comparison is descriptive, not a causal law-quality intervention.

5 Results: law, action, and evaluation

5.1 Executable-law error is not a sufficient proxy for unseen action selection

The longitudinal DeepSeek matrix produced records for 45/45 scheduled cells. Truth evaluation and exact replay both completed 240/240 plan executions. Forty-two cells were uncontaminated and eligible for action metrics; two agent-induced resource/process failures and one session interruption in crystallization remain in the scheduled denominator.

Only 11/42 eligible terminal readouts selected the true Top-1 plan. Mean selected rank was 3.31 of 8 and mean normalized regret was 0.297. The uniform-random rank 4.5 is geometric context, not a no-evidence control, so these outcomes describe post-campaign competence rather than a causal benefit of exploration. Law and action also separated: 30 cells had an inadequate law and wrong action, 11 an inadequate law but correct action, one an adequate law but wrong action, and none an adequate law and correct action. Task means ranged from rank 2.00 for reaction safety to 4.58 for crystallization, and every pairwise arm contrast changed sign under some leave-one-cluster-out omission. We therefore make no pooled arm-level claim (Fig.~4a).

This separation survives continuous and threshold-sensitive analysis. Pooled law MAE correlated weakly with selected rank (Spearman $\rho = -0.073$, cluster-bootstrap 95% interval $[-0.380, 0.256]$) and normalized regret ($\rho = -0.133$, $[-0.452, 0.217]$), while task-specific rank associations reversed sign ($+0.524$, -0.592, and -0.007). Across law-MAE thresholds from 0.05 to 0.30, the adequate subset expanded from 1 to 34 cells but contained only 0 to 9 correct actions. The four-way table is therefore not a single-cutoff artifact: law error does not stably predict unseen-action quality across tasks. A decision-aligned reanalysis of this DeepSeek-v4-flash cohort then executed the last-available law from every frozen cell on the same eight frozen candidate plans. All 45 laws were evaluable, but

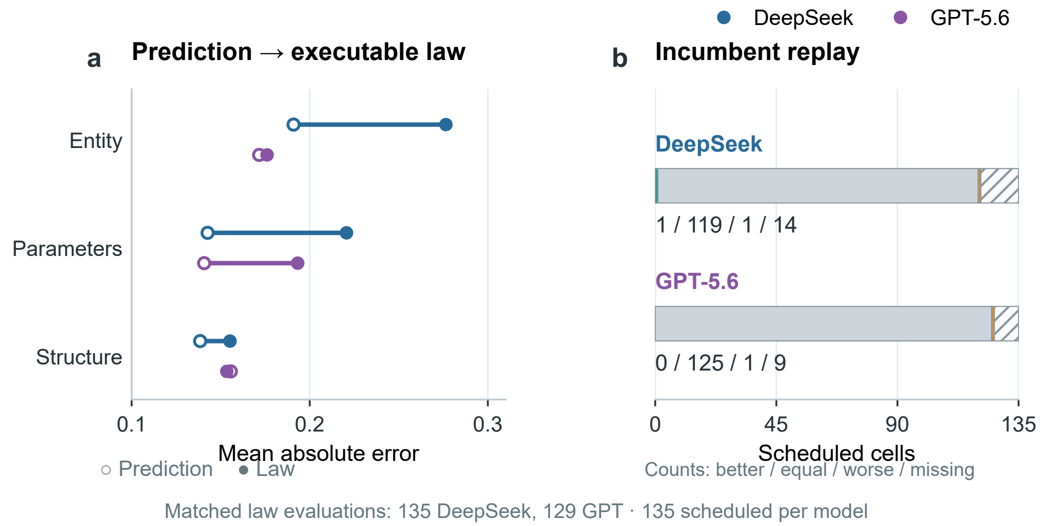

Figure 3: **Lower executable-law error coexists with near-zero incumbent gain.** **a,** Hollow and filled points show final-prediction and executable-law MAE on the same law-evaluable cells, grouped by locus and model; connecting lines show compression differences, not uncertainty. The matched denominators are 135 DeepSeek and 129 GPT laws. **b,** Better/equivalent/worse/unavailable incumbent-replay counts retain all 135 scheduled cells per model; hatching marks unavailable readouts. These are descriptive configuration contrasts and incumbent replays, not causal artifact effects or unseen-action benefits.

their implied choices reached the true Top-1 in 0/45 cells, versus 11/45 when failures were retained for participant action. Among 42 cells with an action ranking, participants followed the law-implied Top-1 in only 12. Three cells without a terminal action ranking still retained an earlier executable law. Participant regret was lower than law-implied regret on average (0.344 versus 0.438), although the direction reversed in crystallization. Because neither law quality nor law following was randomized, this decomposes truth-law error from action-module utilization descriptively rather than estimating a causal law-to-action effect.

5.2 Four-condition strategies expose autonomy and learned-law-only limits

An independent successor scheduled four conditions over all 45 strata per model (360 slots). The all-scheduled failure-aware estimand retains donor, blocked-descendant and recipient failures. Autonomous-minus-no-evidence regret was -0.0913 for DeepSeek-v4-flash (95% task–world-cluster interval $[-0.2124, 0.0388]$) and +0.1102 for GPT-5.6-sol ($[-0.0533, 0.2794]$): directions differed and both intervals crossed zero. GPT's yoked- and learned-law-minus-none estimates were +0.2165 and +0.2459; these include missing donors and system failures, not pure evidence or artifact effects.

Autonomy also changed sign by task: DeepSeek -0.240/-0.406/+0.372 and GPT +0.089/-0.219/+0.460 for electrochemistry/safety/crystallization. Four registered contrasts are descriptive and unadjusted for multiplicity.

Donor-eligible autonomy contrasts (-0.1214/-0.1379; 42/26 strata) are post-treatment sensitivities; equal-task values are -0.0879/-0.0259. Yoked completion was 10/42 and 24/26. The block established no consistent autonomy or learned-law-only advantage (Fig.~4b,c).

5.3 Evaluator-level ranking and decision quality are different estimands

A diagnostic of 16 frozen oracle-control versions found disagreement in both directions: a complete-ranking gate could pass while Top-1 was wrong, or fail while regret was zero. These provider-free controls identify a measurement distinction, not another participant failure. The original stops, all

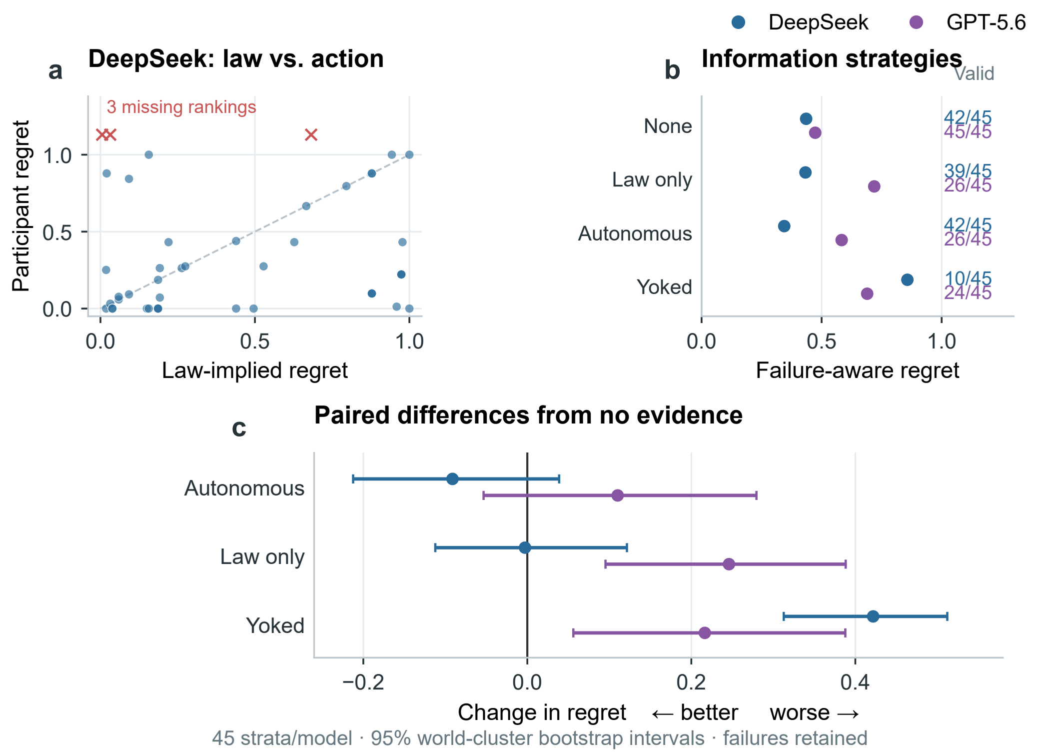

Figure 4: **Law quality and information strategies have distinct decision readouts. a,** Law-implied versus participant regret in the 45-cell DeepSeek matrix; 42 observed choices are dots and three missing rankings are crosses above the plotting range. The diagonal marks equal regret. **b,** Failure-aware strategy means with valid/scheduled counts. **c,** Paired strategy-minus-no-evidence differences with task-stratified world-cluster bootstrap 95% intervals, retaining all 45 strata per model. Both autonomous contrasts cross zero. The strategy block is development evidence; yoked failures prevent a pure acquisition-effect interpretation. Complete-ranking diagnostics remain in the appendix.

unstarted sessions and detailed values remain in the appendix (Table~4); regret and near-optimality directly address selected-action quality.

6 Results: fixed-evidence representation and decision interventions

The factorial intervention holds twelve public experiments per world fixed and crosses two quadratic representations—a source model's law (L) and a public ridge fit (F)—with two decision rules: a fresh same-model agent (A) and a shared deterministic maximizer (X). Five new worlds per task, two model configurations and two source repeats yield 40 source states nested within ten worlds. Sources never see the eight terminal candidates; fresh recipients receive the evidence, candidates and designated law. All 120 provider sessions, 160 condition slots and 200 physical executions with exact replay completed without failure or replacement.

The prespecified primary contrast, F-X minus L-X, was -0.00538 (95% world-bootstrap interval [-0.01630, 0.00061]). Its upper endpoint did not fall below -0.01, so the block does not support the proposed material benefit (Fig.~5). The task means were -0.01087 in electrochemistry and +0.00010 in crystallization; the negative effect was concentrated in one electrochemistry world. Regret uses a fixed utility scale of one, with models and repeats averaged within world. Five worlds per task limit the bootstrap approximation.

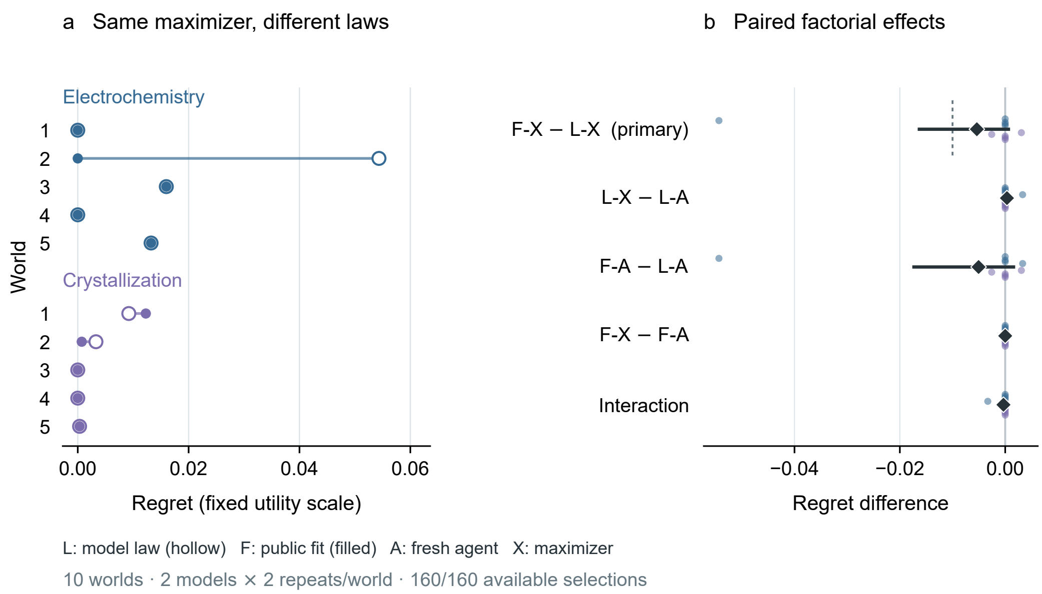

Figure 5: **Fixed-evidence intervention did not establish a material repair benefit. a,** Every world has its own row. Hollow and filled points show model-law and fitted-law regret under the same maximizer, averaged over two models and two repeats; concentric points denote equality. Blue denotes electrochemistry and purple crystallization. **b,** Colored points are individual world effects; black diamonds and lines show means and prespecified intervals (95% primary; 98.75% secondary). The dotted primary reference is -0.01. Interaction is $(R_{\mathrm{F-X}} - R_{\mathrm{L-X}}) - (R_{\mathrm{F-A}} - R_{\mathrm{L-A}})$ in regret. Zero-width intervals describe observed equality, not population equivalence. All 160 selections were available; repeats do not increase the ten-world denominator.

Fresh-agent and maximizer choices agreed in 39/40 model-law pairs and all 40/40 fitted-law pairs. F-X minus F-A was consequently zero in every observed pair; its degenerate bootstrap interval is not a population-equivalence guarantee. Fitted-law selection had mean regret 0.00425, compared with 0.00354 for nearest public evidence and 0.11109 for exact expected uniform-random selection. The fitted method therefore shows no observed advantage over the simple retrieval baseline. These are local response-surface decisions with fixed acquisition. They supply a boundary to general law-use failure, while leaving artifact-only portability and the causes of historical protocol differences unresolved.

7 Related work

Self-driving laboratories and chemistry agents emphasize closed-loop execution, tool use, or endpoint optimization (Felton et al., 2021; Häse et al., 2021; Boiko et al., 2023; Bran et al., 2024). Virtual laboratories and process-control environments provide scalable interaction and safety (Beeler et al., 2024; Bloor et al., 2024; Malik et al., 2026; Chen et al., 2025). Scientific-discovery benchmarks increasingly test iterative experimentation, causal inference, and transferable knowledge (Jansen et al., 2024; Gandhi et al., 2025; Duan et al., 2025; Yang et al., 2026; Batzoglou, 2026). The published ChemWorld platform contributes programmable chemical worlds, transaction semantics and replay (Qiu et al., 2026); this work contributes the intervention and measurement programme for scientific-agent epistemics.

Decision-focused learning distinguishes prediction error from downstream decision loss (Elmachtoub & Grigas, 2022; Wilder et al., 2019). Our additional setting is an agent that acquires evidence under operational constraints and submits a reusable knowledge artifact. This study measures artifact and action outcomes; it does not introduce a decision-focused training algorithm.

Model-discovery systems increasingly combine language models with Bayesian design, symbolic fitting or probabilistic programme search (Murphy, 2026; Wahl et al., 2026; Zheng et al., 2026). We instead ask whether evidence changes the right representation and whether that representation is usable for action. This follows the broader distinction between outcome and process validity (Ríos-García et al., 2026) and the warning that predictive success can coexist with underspecified or shortcut solutions (D'Amour et al., 2022; Geirhos et al., 2020). Causal mediation analysis would require identified interventions on intermediate representations (Imai et al., 2010); we do not infer such mediation from associated law and action readouts. The central contribution is therefore a tiered diagnostic design: controlled initial-model manipulations, conditional packet responses, failure-aware strategy estimates, descriptive law/action decomposition, and a fixed-evidence factorial intervention with separate evaluator qualification.

8 Discussion and limitations

The contribution is a bounded account of experimental knowledge and decision quality. Predictive improvement does not establish selective repair, submitted laws lose different amounts of information, and those laws do not reproduce many participant decisions. The four-condition study estimates information strategies with operational failures retained. It does not isolate a pure evidence-content or experiment-selection effect.

Decision loss, near-optimality and operational availability should accompany prediction error. Continued search and best-minus-first improvement do not by themselves identify feedback learning. Likewise, 11/42 scored Top-1 choices in the original longitudinal cohort do not estimate the benefit of experimentation without its own matched control. Historical rank-gate results concern evaluator design and are not another internal capability failure.

The study covers two fixed simulated model–tool configurations and five worlds per task. B2 is an underidentifying surface with retrospective coding; B3 retains 13 DeepSeek schema failures. Low reasoning is not thinking-off. Donor-eligible analyses are sensitivity-only; fixed order and earlier donors confound configuration with time. These limits preclude model ranking or mediation. Private confirmation and fresh-context transfer to new conditions remain untested.

The fixed-evidence factorial intervention does not establish a material fitted-law benefit. Its near-complete agent/maximizer agreement also limits a general inability-to-deploy-laws account. It does not isolate why the earlier, longer experimental protocol produced greater disagreement: information, function class, decision dimension and interface all differ. Tool-free model fitting versus numerical ridge also bundles arithmetic with artifact construction. A matched-tool control would address that distinction; artifact-only and new-condition tests would address portability. Fit plus argmax is a classical baseline, and no new repair algorithm is claimed.

AI use statement

Generative AI tools assisted software implementation, experiment orchestration, data analysis, visualization code, literature discovery, and drafting and editing of the manuscript. The authors reviewed the study designs and frozen decision rules, inspected retained failures, verified reported denominators and claims against machine-readable records, and tested AI-assisted code before use. The authors take responsibility for the final content, including text, claims, code, and artifacts produced with generative-AI assistance.

Reproducibility statement

The main paper defines the experimental units, interventions, estimands, failure handling, and stop boundaries. The appendix provides the execution and evaluation details needed to reconstruct each evidence layer. Sanitized scheduled-cell records, evaluator-derived metrics, replay and source digests, retained failure classifications, protocol projections, prompts, schemas, and figure source tables are included in the anonymous supplementary package.

Ethics statement

The experiments use simulated chemical environments and no human participants, personal data, or live laboratory actuation. The principal risks are over-interpreting simulated results and treating an evaluator surrogate as the scientific objective; the claim boundaries and retained negative qualification results are reported to mitigate these risks.

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A Appendix

A.1 Environment semantics and intervention construction

An executable world is a stateful transition system with typed operations, measurement functions, finite resources, terminal assays, and an evaluator-only truth surface. A task publishes the operation and instrument contracts while withholding the latent state and registered relations. A scenario binds a world, public task description, resource card, and initial model. Initial-model arms alter one declared information locus without changing the transition process, measurement process, candidate actions, or evaluator truth.

The entity locus changes material identity or ontology. The parametric locus changes a quantitative response model while preserving the declared entities. The structural locus changes the functional form used to describe a mechanism. Aligned and misspecified models are matched in explicitness so that a contrast is not merely information versus no information. The opaque arm exposes the same laboratory interface but withholds the intervened model component. Exploratory, validation, prospective, and private world selections are disjoint by construction.

A.1.1 Task–locus roster and world splits

The nine prospective task–locus combinations are fixed before participant execution. The entity intervention swaps matched nominal descriptor rows while leaving physical identity fixed. Parametric and structural interventions replace only the declared response component; all other task text, tools, resources, queries, and world state remain matched.

Table 2: **Nine prospective task–locus combinations.** “Registered target” names the component manipulated in the initial model, not information revealed by the evaluator.

Locus	Task	Registered target	Held-out diagnostic
Entity	Electrochemical conversion	solvent identity descriptors	conversion/selectivity response
Entity	Reaction–crystallization	solvent identity descriptors	reaction and crystal outcomes
Entity	Reaction–distillation	solvent identity descriptors	reaction and separation outcomes
Entity	Partition discovery	extractant identity descriptors	phase-partition outcomes
Entity	Reaction safety	catalyst identity descriptors	yield, kinetics, and risk
Parametric	Electrochemical conversion	quantitative potential-response direction	peak-and-collapse profile
Parametric	Reaction safety	quantitative catalyst/thermal response	productive versus hazardous regime
Structural	Partition discovery	linear versus power-form phase relation	registered exponent 1.75
Structural	Reaction–crystallization	pathway/topology and accumulation form	multistage outcome surface

Development and qualification use seeds 0–4 and never enter the prospective denominator. Public worlds are five task-specific SHA-256-derived identities per task; their values are fixed in the released configuration. The private split is bound by a commitment but was not unsealed or executed. Matched-evidence and action studies use separately frozen public surfaces. Sessions sharing a task–world but differing in prior arm are repeated measurements within one inference cluster.

A.1.2 Metrics and registered correction gates

For prediction or executable law f, truth y, registered query–metric pairs $j = 1, \ldots, J$, and fixed positive metric scale s_j (one for bounded metrics unless overridden), normalized error is

$$E(f) = \frac{1}{J} \sum_{j=1}^{J} \min\left(\frac{|f_j - y_j|}{s_j}, 1 \right).$$

For candidate scores q_i, selected candidate a, best $q_{\max}$, and worst $q_{\min}$, normalized regret is $(q_{\max} - q_a)/(q_{\max} - q_{\min})$, with zero assigned when the packet range is numerically zero. A missing or invalid terminal ranking receives failure-aware regret one and Top-1 zero. Near-optimal selection means raw regret at most 0.01; pairwise ordering excludes truth pairs separated by less than 0.01.

Let $I_a = E_{a,0} - E_{a,K}$ and $C = I_{\text{mis}} - I_{\text{aligned}}$. The entity gate is a three-component intersection–union test at one-sided $\alpha = 0.05$: the lower bound for C must exceed zero, misspecified improvement must exceed zero, and aligned improvement must exceed the non-inferiority margin -0.05. Parametric and structural gates use equal-task fixed-effect estimates of C, require the one-sided lower bound above zero and positive direction in every included task. For an incomplete cell, confirmatory bounds use the adverse interval $[-1, +1]$ for arm improvement; the corresponding contrast can therefore span $[-2, +2]$. Last-observation-carried-forward and zero improvement are observed-point sensitivities only and do not generate the confirmatory p values.

A.2 Persistent execution and checkpoints

Each participant cell is one persistent agent process. At every turn, the agent may call only the public laboratory tools. The host validates schemas and preconditions, applies resource accounting, commits successful transactions, and returns public observations. It does not choose or repair an experiment on the participant's behalf. Entity, parametric, and structural campaigns permit eight, ten, and twelve complete experiments, respectively. Checkpoints occur at experiments 0/2/4/6/8, 0/2/4/7/10, and 0/3/6/9/12.

Every checkpoint records model-reliability judgements, query predictions, uncertainty, public evidence references, a typed executable law, and the intended next experiment. Checkpoint repair is part of the evaluated interface burden. All eventually yielded valid records, but rejected attempts remain in the process log and resource accounting.

A.2.1 System, prompt, tool, and budget contract

`DeepSeek-v4-flash` with `reasoning_effort=high` supplies the primary DeepSeek C2, matched-evidence, B3-successor, and action surfaces; the same model at `low` supplies only the completed B2 reasoning-budget ablation. `GPT-5.6-sol` with `reasoning_effort=medium` supplies the matched GPT replications. Both run through the Codex session harness; "Codex" therefore denotes execution infrastructure, not a third participant model. Provider aliases, run dates, receipt counts, and usage fields remain in the retained internal records rather than the anonymous supplement. The supplement retains sanitized model/configuration labels, scheduled completion denominators, and derived resource summaries. Temperature is not exposed or set by the harness. Request timeout is 1,200 s; finalization timeout is 600 s; progress is emitted every 30 s.

Each C2 cell receives one task/scenario prompt, the same public operation schemas, the arm-specific initial model, and no evaluator truth. A persistent transcript spans the complete campaign. The fixed 8/10/12-experiment budgets allow at most 96/110/168 operation attempts, 480k/600k/720k uncached input tokens, 128k/160k/192k output tokens, and 7,200/7,200/9,000 s wall time, respectively. Each cell uses at most two provider calls within the same session; infrastructure-only resume is capped once, while any persisted scientific trajectory forbids replacement. Matched-evidence assays use exactly two same-thread turns: a pre-evidence submission and a post-packet submission. The supplement includes the exact static system prompts, selected prompt-generating function sources, public protocol projections, representative strict response schemas, and the executable-law contract. The package also contains all 45 sanitized B2 public summaries, the retrospective expression-coding function, three-configuration all-arm counts, and the participant-visible B2 identifiability audit.

A.2.2 Executable-law schema and ActionPlan construction

A law contains one metric equation for every required metric, an intercept, identity or logistic link, output bounds, evidence references, confidence, applicability, and limitations. Each metric permits at most 64 terms drawn from linear, quadratic, cubic, pairwise interaction, categorical indicator, and category-conditional linear/quadratic/cubic bases. Terms may reference only the task's registered feature IDs; evidence references may name only participant-visible assays. The production parser and executor score the law directly, so prose that cannot be parsed does not silently receive a law score.

Terminal candidates are eight complete, compiler-valid ActionPlans selected from a registered 16-query public-feature roster without reading outcomes. Each packet declares every ordered operation and parameter, fresh initial state, measurement positions, terminal assay, omitted optional operation, and objective. The other eight rows form the disjoint prediction/oracle-fit roster. Public-plan, truth-plan, and replay hashes must agree exactly. Candidate outcomes and evaluator ranks remain hidden until scoring.

A.3 Evaluator-owned prediction and law assays

Entity checkpoints contain four prespecified counterfactual queries; parametric and structural checkpoints contain sixteen. The evaluator executes each unique task–world query independently, never returns its truth to the participant, and shares truth across prior arms and checkpoints. The primary readout is mean normalized absolute error across registered query–metric pairs. The executed cohort contains 420/420 truth executions, 1,620 query–metric truth values, 675/675 checkpoints, 6,300 predictions, and 24,300 scored values.

Final typed laws are executed on the same registered coordinates. Schema validity, coordinate coverage, truth-normalized error, change from pre-evidence and effective-final checkpoints, and consistency with explicit final predictions are retained separately. This assay establishes executability and in-domain compression only. Reusability would require new coordinates or transfer to a context-reset process.

The matched GPT-5.6-sol evaluator uses the identical 45-cluster, nine-task, three-arm C2 surface. DeepSeek-v4-flash and GPT-5.6-sol have 121/135 and 126/135 completed cells. Their exact evaluator denominators are 675/675 versus 669/675 scored checkpoints, 135/135 versus 129/135 executable laws, and 121/135 versus 126/135 cells with evaluable blind gain. Overall mean prediction improvement is 0.1198/0.1329, final prediction error is 0.1685/0.1614, law MAE is 0.2371/0.1753, law compression loss is 0.0686/0.0142, and blind gain is -0.0010/-0.0001. GPT-minus-DeepSeek paired descriptive differences are +0.0131 for prediction improvement, -0.0071 for final error, -0.0648 for law MAE, and -0.0579 for compression loss. Both models fail all three registered selective-correction gates.

A.4 Typed-law capacity and distillation controls

To distinguish participant compression failure from an underpowered output schema, we refitted each of the 135 complete final prediction vectors with legal identity-link laws over the registered feature coordinates and allowed basis functions. A full-basis fit used up to 64 terms per metric. A term-matched fit used exactly the participant's submitted term budget for each metric. A leave-one-query-out fit withheld each registered query in turn and predicted it from all remaining queries. Every fitted law was parsed and executed by the production law evaluator; independently computed design-matrix predictions had to agree with executor output within 10^{-10}.

The participant laws differed from their own final predictions by mean MAE 0.1539. Full-basis fits reproduced 135/135 final prediction states with mean MAE 4.25×10^{-13}; term-matched fits reduced the error to 0.0114 and leave-one-query-out fits reached 0.0788. The control uses the same registered coordinate domain and therefore tests representation capacity and distillation, not global mechanistic identification or transfer.

A.5 Failure-aware inference

The unit of prospective inference is the independently selected task–world cluster, not a checkpoint, query, experiment, or replay. The three prior arms within a cluster are matched. Locus-specific selective-correction contrasts are computed first and are not pooled across different intervention semantics. Failed scientific cells remain assigned. Confirmatory correction bounds assign an incomplete post-outcome the adverse improvement range $[-1, +1]$; last-observation-carried-forward and zero improvement are observed-point sensitivities only. Infrastructure recovery is allowed only when no scientific trajectory has persisted and only under the fixed attempt limit.

The matched-evidence B2 contrast has five independent worlds. Its interval and exact one-sided sign-flip result are descriptive because the block was designed to localize a transition, not to establish a population-level arm effect. Recipe divergence is likewise a manipulation check: without repeated same-arm sessions, provider stochasticity cannot be separated from intervention sensitivity at exact-recipe resolution.

DeepSeek and GPT-5.6-sol medium used identical matched-evidence worlds, arms, evidence packets, queries, and scoring rules. Each completed 15/15 A-P and 15/15 A-S B2 formal sessions with no failed cells; canaries were excluded. A-P misspecified-minus-aligned update contrasts were 0.0309 and 0.0602, with 3/5 and 5/5 positive worlds. A-S B2 contrasts were 0.0645 and 0.0915, with 3/5 and

4/5 positive worlds; misspecified exact 1.75-law expression was 0/5 in both models. Configurations are reported separately, and no model-superiority test is performed.

The B2 exact-law counts are retrospective keyword coding of public `model_summary` and `evidence_assessment` fields; no typed family or exponent field was preregistered. A later participant-visible audit found that evidence and scoring fixed one nominal solvent/extractant pair, did not expose the base partition coefficient, and admitted an exact alias between the 1.75-power law and a free-coefficient linear law (coefficient multiplier 3.13588). A constant endpoint baseline obtained mean scoring MAE 0.00649, while the aligned DeepSeek-high exact-law positive control was only 1/5 and failed its readout criterion. B2 therefore supports conditional low post-packet error and lack of stable exact-law expression on an underidentifying surface, not a participant-level structural-identification failure. The separate B3 control below supplies that identifiable test.

The reasoning-budget ablation kept the DeepSeek model, Codex harness, prompts, schemas, public packets, worlds, and evaluator fixed while changing only `reasoning_effort` from high to low. It is not provider-level thinking-off, which would require a different direct-controller harness. The B2 interface canary passed 3/3 and the formal block completed 15/15 sessions, 30/30 turns, 15/15 same-thread continuations, and 360/360 pre plus 360/360 post scoring terms, with no failed session or infrastructure predecessor. Mean post errors were 0.0067/0.0069/0.0069 for opaque/aligned/misspecified. The registered contrast was -0.0405, with 2/5 positive worlds, exact one-sided $p = 0.8125$, and descriptive interval [-0.1559, 0.0749]. Misspecified exact 1.75-law expression remained 0/5; all five misspecified post errors were at most 0.02. Aligned exact-law expression was 1/5 for DeepSeek high, 0/5 for GPT medium, and 2/5 for DeepSeek low, reinforcing that the free-text expression readout itself lacks a strong positive control.

Provider-reported DeepSeek reasoning output was 506,637 tokens at high effort and 400,639 at low effort on the matched B2 block, a 20.9% reduction. These resource fields are descriptive within the DeepSeek provider and are not compared with GPT accounting. The separate low-effort A-P canary was platform-defective: progress events existed, but terminal cell receipts and a canary summary did not. It is retained as an unscored partial; A-P low formal execution remained 0/15 and supplies no effect estimate.

A.6 Reference-fitter-identifiable law and action control

The independent B3 control used five frozen public structural-partition worlds, the same opaque, aligned, and misspecified prior arms, and two independent fresh GPT-5.6-sol medium sessions per arm and world. Provider-free development qualification selected eight evidence rows spanning four nominal pairs and a disjoint eight-query scoring/action roster. Public truth was executed before any participant call. All five worlds entered family, exponent, prediction, Top-1, rank, and regret denominators; action gain was scored only in the three worlds whose best candidate exceeded the visible evidence incumbent by at least 0.02. The evidence and scoring rosters, five worlds, three arms, two replicates, 0.10 exponent tolerance, and 0.02 opportunity threshold were frozen before participant execution.

A three-session canary tested only two-turn schema validity, packet identity, same-thread continuity, typed law fields, the exact scoring denominator, and selection of an unexecuted candidate. It passed 3/3 and did not enter the formal scientific denominator. The full 30-cell block then completed 30/30 with no failed cells or replacement. All sessions preserved same-thread continuity and returned 60/60 completed formal turn receipts. The canary and formal blocks together used 33 provider session attempts, 66/66 completed turn receipts, zero retries, zero tool events, and zero participant physical experiments. One formal post turn recorded a transient provider error event before returning a completed receipt; the event remains in the resource ledger and the cell remains completed.

Joint structural recovery required both the registered power family and exponent error at most 0.10. Counts for opaque, aligned, and misspecified arms were 0/10, 5/10, and 0/10. The corresponding mean post-evidence errors were 0.0367, 0.0215, and 0.0378. The aligned world-mean exponent error was lower than each comparator in all five worlds. Top-1 counts were 0/10, 2/10, and 0/10; both Top-1 choices came from one action-ineligible world. None of the 18 action-eligible cells reached gain 0.02, including neither of the two action-eligible cells with joint structural recovery. A matched DeepSeek B3 successor then started from its first cell on the same scientific surface and retained every scheduled outcome. It completed 17/30 cells; 13/30 were participant-schema failures.

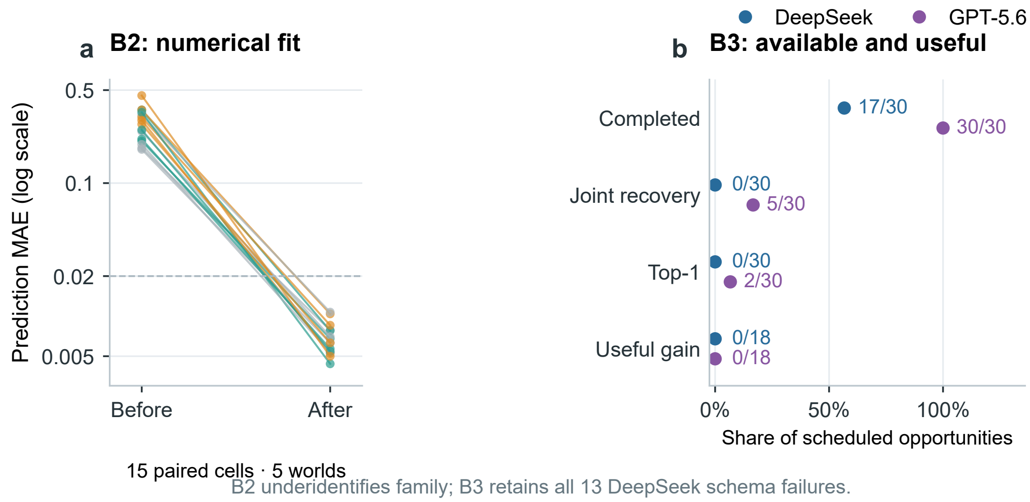

Figure 6: **Numerical fit and structural recovery require different tests.** **a,** All 15 DeepSeek-high B2 cells before and after the same packet, shown on a log scale; gray/teal/orange denote opaque/aligned/misindexed arms. All post-packet errors are below 0.02, but the one-pair surface underidentifies family; low error is not structural recovery. **b,** B3 completion, joint family–exponent recovery, Top-1, and useful gain retain all scheduled opportunities. Counts distinguish 30 sessions from 18 gain opportunities per model, including 13 DeepSeek schema failures. Configuration contrasts are descriptive.

Failure-aware joint recovery was 0/30, Top-1 was 0/30, and mean regret was 0.9579; completed-cell post MAE was 0.0928. The corresponding GPT-5.6-sol values were 5/30, 2/30, 0.7594, and 0.0320. Useful-gain success was 0/13 versus 0/18 among completed evaluable opportunities and 0/18 for both models on the fixed scheduled-opportunity denominator, with failures and unavailable rows counted as zero. The matched comparison is descriptive because model configuration was not randomized and schema-failure patterns differed.

A.7 Cross-model successor denominators

Historical stopped partials and stop boundaries remain immutable. They are not continued, pooled, or converted into successful cells. Each successor instead starts at its first scheduled unit with a complete failure-aware denominator.

The GPT-5.6-sol C2 successor scheduled all 135 task–world–prior cells and terminated with 126 completed, 3 failed, and 6 right-censored sessions, completing 1,253/1,260 participant experiments. Its provider-free evaluator completed 420/420 truth executions, scored 669/675 checkpoints and 129/135 laws, and launched 756/810 scheduled blind executions; 54 remained unstarted. The matched 135-cell DeepSeek/GPT analysis uses task–world cluster bootstrap intervals and never interprets model differences as provider effects.

The DeepSeek B3 successor similarly schedules 30 cells and retains 17 completed plus 13 participant-schema failures. Its historical canary is excluded rather than spliced into the new denominator. The GPT B3 cohort remains the independent completed 30-cell block described above.

The four-condition action successor schedules 45 strata by four conditions for each model: 180 slots per model and 360 total. DeepSeek reuses 45 immutable autonomous donors, of which 42 are eligible; GPT-5.6-sol creates a separate 45-donor cohort, of which 26 are eligible. No-evidence is scheduled for all strata, while yoked and learned-law recipients require the corresponding donor. Missing-donor descendants remain `not_started_due_to_missing_donor`. The primary analysis is the model-specific all-scheduled failure-aware strategy estimand over all 45 strata. DeepSeek/GPT autonomy-minus-no-evidence estimates are -0.0913 and +0.1102 with clustered intervals [-0.2124, 0.0388] and [-0.0533, 0.2794]. Donor-eligible estimates (-0.1214/-0.1379) are availability-conditioned sensitivities because eligibility is post-treatment; equal-task sensitivities are -0.0879 for DeepSeek and -0.0259 for GPT. The cross-model common donor-eligible population contains 26 strata from 13 task–world clusters. Yoked completion is 10/42 for DeepSeek and 24/26 for GPT, so autonomous–yoked contrasts mix scientific action quality with recipient-system failure and are not pure experiment-selection effects. The autonomous-minus-no-evidence task estimates were -0.2398/-0.4060/+0.3720 for DeepSeek and +0.0891/-0.2191/+0.4605 for GPT over electrochemistry/safety/crystallization, respectively. Thus the pooled direction masks strong task heterogeneity. All four registered contrasts and their bootstrap intervals are descriptive and were not multiplicity-adjusted. Conditions followed fixed manifest and dependency order rather than randomization or counterbalancing; DeepSeek autonomous donors also predated their recipients. Provider/time drift and condition-order effects are therefore not separable.

A.7.1 Recovery and composite provenance

Platform repair never changed a scientific question, coverage set, metric, or gate after observing an outcome. Affected blocks restarted from their first affected unit; discarded or superseded records remain in the failure timeline and do not enter the current estimate.

Table 3: **Defect and recovery provenance for current publication surfaces.**

Surface	Retained incident	Recovery boundary	Current inclusion
DeepSeek C2	one 15-cell shard affected by runtime/evaluator defects	whole shard restarted; evaluator rerun over the full composite	120 unaffected cells plus one 15-cell replacement shard
GPT C2	no inherited stopped cell	new 135-cell block from cell 1	all 135 scheduled outcomes
DeepSeek B3	historical three-cell canary failed schema	independent 30-cell successor from cell 1	17 complete plus 13 schema failures
Four-condition yoked	duplicate field deletion caused recipient KeyError	entire yoked condition restarted from its first admitted unit	recovered condition only; original failures retained in timeline
Oracle grids	path-length and evidence-reference limits	identical scientific block restarted after platform repair	only complete version-specific qualification blocks

The four-condition report remains a prospective development experiment rather than confirmatory evidence; the reanalysis does not rename it formal evidence. The DeepSeek C2 current composite is reported as a composite, not as one uninterrupted launch. Exact source bindings and excluded incident counts are in the supplement.

A.8 Complete-plan action assay

In each open-action cell, a persistent agent performs twelve autonomous experiments and commits checkpoints after 0, 3, 6, 9, and 12 experiments. The host then reveals eight unseen ActionPlans. Each plan specifies ordered operations, all submitted parameters, initial-state assumptions, intermediate measurement positions, terminal assay, and declared omitted operations. Candidate outcomes, ranks, and other-arm evidence stay hidden. The evaluator verifies identity of disclosed, truth-evaluated, and replayed plans.

The primary endpoint is normalized regret of the selected plan within its task–world candidate set. Selected rank, Top-1, pairwise ordering, and law adequacy are secondary or diagnostic. Forty-five cells were scheduled across three tasks, five worlds, and three initial-model arms. Forty-two were eligible. The two agent-induced resource/process failures and one provider interruption are retained as failures in the scheduled denominator and are not replaced. An independent repair is reported only as sensitivity evidence because it follows a new trajectory.

The continuous law–action analysis used only those same 42 originally eligible cells; the three failures stayed in the 45-cell scheduled denominator and were not imputed. Pearson and Spearman associations were computed pooled and by task. Uncertainty resampled the 15 frozen task–world clusters, retaining all eligible arms within a sampled cluster, with 10,000 bootstrap replicates and seed 20260827. The pooled Spearman coefficients were -0.073 for law MAE versus selected rank and -0.133 for law MAE versus normalized regret, with cluster-bootstrap 95% intervals $[-0.380, 0.256]$ and $[-0.452, 0.217]$. A prespecified sensitivity table swept law-MAE thresholds 0.05, 0.075, 0.10, 0.15, 0.20, 0.25, and 0.30 without selecting a cutoff from action outcomes.

A decision-aligned reanalysis used the last-available executable law from all 45 frozen DeepSeek-v4-flash longitudinal cells and reconstructed the same eight candidate feature packets without new participant, truth, or physics execution. Three cells had no terminal action ranking but retained an earlier executable law. For each law, truth-law error is the normalized regret of its implied Top-1. Action-utilization delta is participant regret minus law-implied regret. All 45 laws executed; none implied the true Top-1, whereas participant action selected Top-1 in 11/45 scheduled cells. Among 42 cells with a valid participant ranking, the participant followed the law-implied Top-1 in 12. Mean law-implied and participant regret were 0.438 and 0.344. This is a descriptive decomposition: neither law quality nor law use was randomized.

A.9 ORACLE-CONTROL QUALIFICATION

The unexecuted causal follow-up crossed three tasks, five worlds, three priors, and five conditions: no evidence, stepwise yoked evidence, autonomous exploration, learned law in a fresh context, and a provider-free oracle law. The planned denominator was 225 participant sessions, 45 autonomous donors, and 540 donor experiments. Participant execution required every fresh task–world oracle to pass candidate opportunity checks and to attain Spearman rank correlation at least 0.80 over the eight candidate plans using fit data disjoint from candidate outcomes.

The 96-query qualification stopped after eight of fifteen clusters. It completed 896/896 truth executions and exact replays, with eight candidate gates and seven oracle rank gates passing. The retained crystallization failure had $\rho = 0.738095$, Top-1 disagreement, and zero fit/candidate overlap. No operational canary or participant session began.

This failure initiated a transparent development sequence rather than a retrospective threshold change. Version 0.2 first failed on a fresh electrochemical world at $\rho = 0.785714$. Version 0.3 added exact typed-law distillation but failed on a fresh crystallization world at $\rho = 0.595238$. Version 0.4 fixed an ExtraTrees predictor: it replayed all 25 exposed construction worlds successfully, then stopped after five fresh units when the fifth reached only $\rho = 0.785714$ (four pass, one fail, ten unstarted). Each version used a newly held-out fresh surface, stopped at its first registered failure, read no candidate outcomes during fitting, and made zero provider calls. These attempts are not pooled into a success rate and do not modify the original stop decision.

The 320-query construction held the fitted ExtraTrees family fixed while expanding coverage to 64 global and 256 candidate-neighborhood queries. A platform-defective partial caused by an evidence-reference schema limit was retained and not reused. After the contract repair, an independent construction run completed 2,352/2,352 truth executions and replays across seven exposed units, passing all seven and repairing four historical failures. The first prospective world then completed 336/336 executions and replays but failed at $\rho = 0.714286$; the remaining fourteen were not started. Candidate outcomes were never read during fitting.

The gate-alignment analysis added no execution. It re-read sixteen frozen unit versions and exactly reproduced their original Spearman and Top-1 values. Its action endpoints were Top-1, normalized regret, selection within 0.01 of optimum, and near-tie-aware pair ordering. It did not alter any historical result, threshold, or stop decision.

For eight candidates without ties, $\rho = 1 - \sum_i d_i^2/84$. Thus the retained 96-query failure at $\rho = 0.738095$ has $\sum_i d_i^2 = 22$, whereas the fresh 320-query failure at $\rho = 0.714286$ has 24. The scalar measures total displacement, not whether the best action moved.

Table 4: **Complete-ranking gates and action endpoints are different estimands.**

Frozen block	n	Rank pass	Top-1	$\leq$ 0.01 regret
Fresh 96-query	8	7	1	3
Exposed 320-query	7	7	4	6
Fresh 320-query	1	0	1	1

A.10 INDEPENDENT-WORLD REPRESENTATION-BY-DECISION PROTOCOL

The factorial block fixes public evidence while replacing either its explicit representation or the decision rule. Each of two task families has five new public-test worlds. Two fixed model configurations each produce two independently sampled source artifacts per world, giving 40 source states nested within ten task–world clusters. Every source state has four conditions: model law with fresh agent selection (L-A), the same law with a deterministic maximizer (L-X), public-data fit with fresh agent selection (F-A), and the same fit with the maximizer (F-X). The scheduled design comprises 40 source and 80 recipient sessions, 160 condition slots, 120 public evidence experiments and 80 hidden candidate evaluations, with exact replay of each physical execution. This block uses fixed experiment acquisition.

Each world shares twelve evidence points and eight outcome-hidden candidate plans across models and repeats. Independently seeded Latin hypercubes fix the two-dimensional coordinates before outcomes are observed. Electrochemistry varies controlled potential over 0.65–1.65 V and current over 20–120 mA. Its fixed probe uses 1.18 V, 70 mA and 630 s; controlled duration is 3540 s, reagent amount 0.004 mol, electrolyte profile 2 and solvent 1. The complete LHS is rejected before execution if any point violates minimum probe changes of 0.02 V or 1 mA. Crystallization varies reaction temperature over 350–405 K and crystallization temperature over 275–295 K. Catalyst 0 at 0.000315 mol, solvent 1, reagent 0.015 mol, stirring 675 rpm, reaction duration 3600 s, seed mass 0.008 g and crystallization duration 7200 s remain fixed. Both tasks share the same normalized design and disclose the complete executable plans.

L and F use the same unclipped quadratic basis $(1, x, y, x^2, xy, y^2)$ in linearly normalized coordinates. F is ridge regression on the twelve public utilities, with penalty 10^{-6} and an unpenalized intercept. Source models see evidence before candidates are revealed and return six finite coefficients. Fresh recipients receive evidence, candidates and one artifact, and return only a candidate identifier. Calls are tool-free, capped at 600 s each, with no repair turn or replacement. The request for 2,048 output tokens does not cap hidden reasoning; actual usage is recorded. Source states have a fixed outcome-blind permutation, and each world/model uses one L-first and one F-first decision repeat. The maximizer breaks exact ties by candidate order. All choices are sealed before candidate utilities are loaded for analysis.

Candidate utility is a single keyed-noise measured score in $[0, 1]$. M1 regret uses the fixed utility scale one; earlier action assays retain their registered range normalization. Missing or invalid choices receive regret one in the scheduled primary analysis; completed-only values retain separate denominators. Near-optimality means regret at most 0.01. The primary contrast is F-X minus L-X. Differences are first averaged over the four model/repeat states within world, then equally across worlds within task and across tasks. A task-stratified world bootstrap uses 20,000 percentile replicates. Its two-sided 95% upper endpoint must be below -0.01 to support the prespecified material benefit. The four secondary contrasts (L-X minus L-A, F-A minus L-A, F-X minus F-A, and their factorial interaction) use 98.75% marginal intervals, a Bonferroni adjustment for four comparisons. Five worlds per task give approximate small-sample intervals; model repeats do not add independent worlds.

The nearest-public-observation baseline uses normalized Euclidean distance and row-order ties; the random baseline is its exact uniform-candidate expected regret. Both reuse the same data and count once per world. Missing source artifacts block the dependent L conditions while F continues. Physical, semantic or replay failure stops dependent provider work; forbidden tools and missing or reused session identity invalidate the remaining block. Interrupted started slots are retained without retries. Physical CPU/wall includes replay; recipe-resource totals count primary execution once. The protocol replaces explicit artifacts and decision computation, and does not identify internal belief mediation or artifact-only transfer: recipients still receive the original evidence.

A.10.1 FACTORIAL OUTCOMES AND RESOURCES

All 40 source states, 120 sessions and 160 conditions completed; there were no failed, blocked, replaced or unstarted slots. Scheduled and completed-only losses coincide. Mean regrets in L-A/L-X/F-A/F-X order were 0.01436/0.01502/0.00425/0.00425 for DeepSeek and 0.00425/0.00425/0.00425/0.00425 for GPT. These configuration-specific summaries are descriptive. F-X selected the exact best measured candidate in 20/40 states and a candidate within 0.01 in 28/40;

the replicated fit copies still represent only ten worlds. Candidate MAE was 0.05028 for DeepSeek laws, 0.04832 for GPT laws and 0.04459 for the public fit, all finite. Better average prediction error thus did not establish a material decision improvement.

Table 5: **Prespecified factorial contrasts.** Differences use the fixed regret scale. The primary material threshold is -0.01. Secondary intervals adjust for four comparisons.

Contrast	Mean difference	Interval
F-X minus L-X (primary)	-0.005384	95% [-0.016303, 0.000614]
L-X minus L-A	+0.000331	98.75% [0.000000, 0.001323]
F-A minus L-A	-0.005053	98.75% [-0.017329, 0.001582]
F-X minus F-A	0.000000	98.75% [0.000000, 0.000000]
Interaction	-0.000331	98.75% [-0.001323, 0.000000]

The ten-world average nearest-evidence regret was 0.00354, versus 0.00425 for F-X and 0.11109 for exact expected uniform-random choice. These prespecified baselines reuse the same evidence and candidate evaluations; the comparison supplies no novel-algorithm advantage. The F-A/F-X bootstrap interval collapses to zero because all observed paired choices agree. It does not establish exact equivalence for unobserved worlds or model samples.

Physical execution and replay consumed 1,780.8 s wall time and 1,768.3 s CPU. The 120 public evidence executions account for 1,068.6 s wall time; 80 hidden evaluations account for 712.3 s. Recipe resources count primary executions once: 2,300 operations, 54.8 measurement units, 1,497,000 simulated recipe seconds and 1.9 mol reagent. Provider calls used 8,859.5 s wall time, 1,623,791 input tokens (450,816 cached) and 942,258 output tokens (939,441 reasoning). All 120 calls report input/output usage; reasoning is included in output and cache in input. Source generation used 7,404.1 s versus 1,455.4 s for decisions. These costs describe this block, without extrapolating provider billing or virtual resource use to laboratory expenditure.

A.11 Additional scope boundaries

The study covers simulated chemistry and two fixed agent–tool configurations, not laboratory fidelity or universal chemical coverage. C2 and B3 have matched scheduled surfaces; A-P/B2 have complete separate denominators plus a B2-only DeepSeek-low ablation. Differential failures and non-random provider assignment prohibit model ordering. The public cohort is not private confirmation, and high-fidelity artifact portability remains untested. The autonomous open-action cohort lacks a same-agent no-evidence baseline; the oracle-free successor adds four information strategies, but donor/recipient failures limit mediation claims. The original five-condition oracle cohort remains unexecuted.